Clustering Analysis
Implement K-Means clustering on Iris.csv dataset. Determine the number of clusters using the elbow method. Dataset Link: https://www.kaggle.com/datasets/uciml/iris

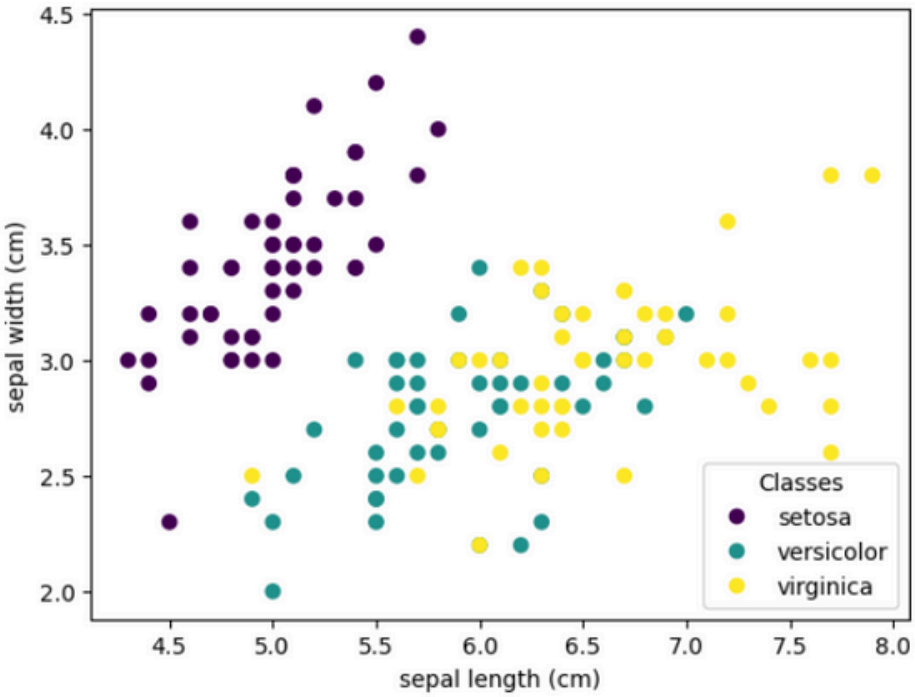
Code :

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

iris = load_iris()

dir(iris)
```

```
['DESCR',
 'data',
 'data_module',
 'feature_names',
 'filename',
 'frame',
 'target',
 'target_names']
```

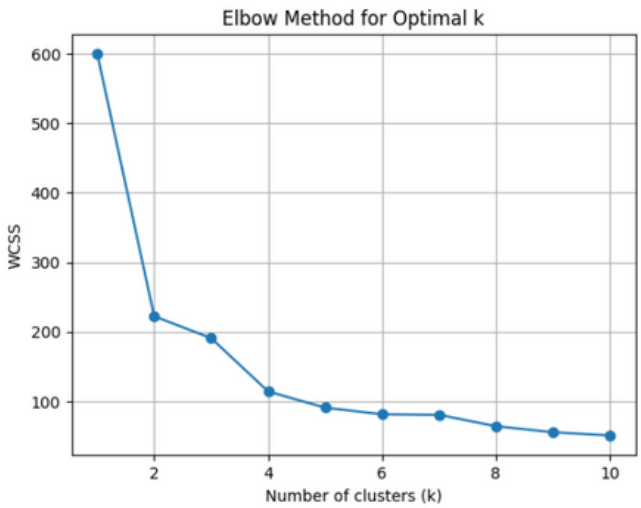
Before clustering

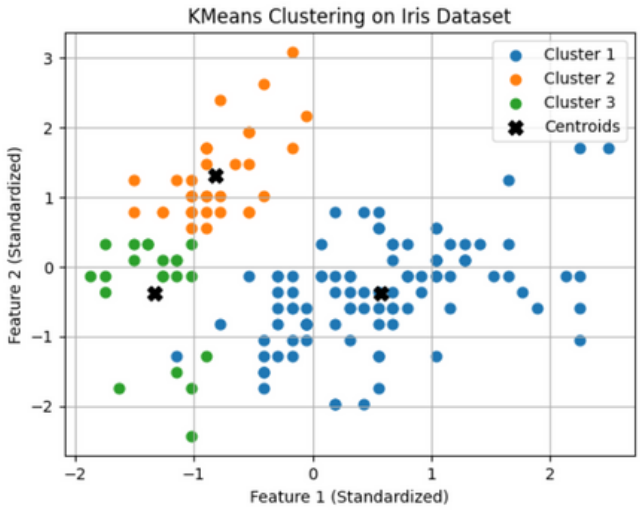
```
import matplotlib.pyplot as plt

_, ax = plt.subplots()
scatter = ax.scatter(iris.data[:, 0], iris.data[:, 1], c=iris.target)
ax.set(xlabel=iris.feature_names[0], ylabel=iris.feature_names[1])
_ = ax.legend(
    scatter.legend_elements()[0], iris.target_names, loc="lower right", title="Classes"
)
```



Applying Elbow chart method to identify the accurate no. of groups

```
X = iris.data  # Only features (4 features: sepal/petal length/width)

# Optional: Standardize the features for better clustering
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

wcss = []  # Within-cluster sum of squares

# Try K from 1 to 10
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot the elbow graph
plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow Method for Optimal k')
plt.xlabel('Number of clusters (k)')
plt.ylabel('WCSS')
plt.grid(True)
plt.show()
```




```
optimal_k = 3
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)

plt.scatter(X_scaled[y_kmeans == 0, 0], X_scaled[y_kmeans == 0, 1], s=40, label='Cluster 1')
plt.scatter(X_scaled[y_kmeans == 1, 0], X_scaled[y_kmeans == 1, 1], s=40, label='Cluster 2')
plt.scatter(X_scaled[y_kmeans == 2, 0], X_scaled[y_kmeans == 2, 1], s=40, label='Cluster 3')

# Smaller centroid markers too
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1],
            s=80, c='black', marker='X', label='Centroids')

plt.title('KMeans Clustering on Iris Dataset')
plt.xlabel('Feature 1 (Standardized)')
plt.ylabel('Feature 2 (Standardized)')
plt.legend()
plt.grid(True)
plt.show()
```